MILITARY SPECIFICATION SHEET

GENERATOR SYSTEM, 20 KVA, 400 HERTZ ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS TYPE

THIS SPECIFICATION HAS BEEN APPROVED BY THE NAVAL AIR SYSTEMS COMMAND, DEPARTMENT OF THE NAVY. THE COMPLETE REQUIREMENTS FOR PROCURING THE GENERATOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF MIL-G-21480.

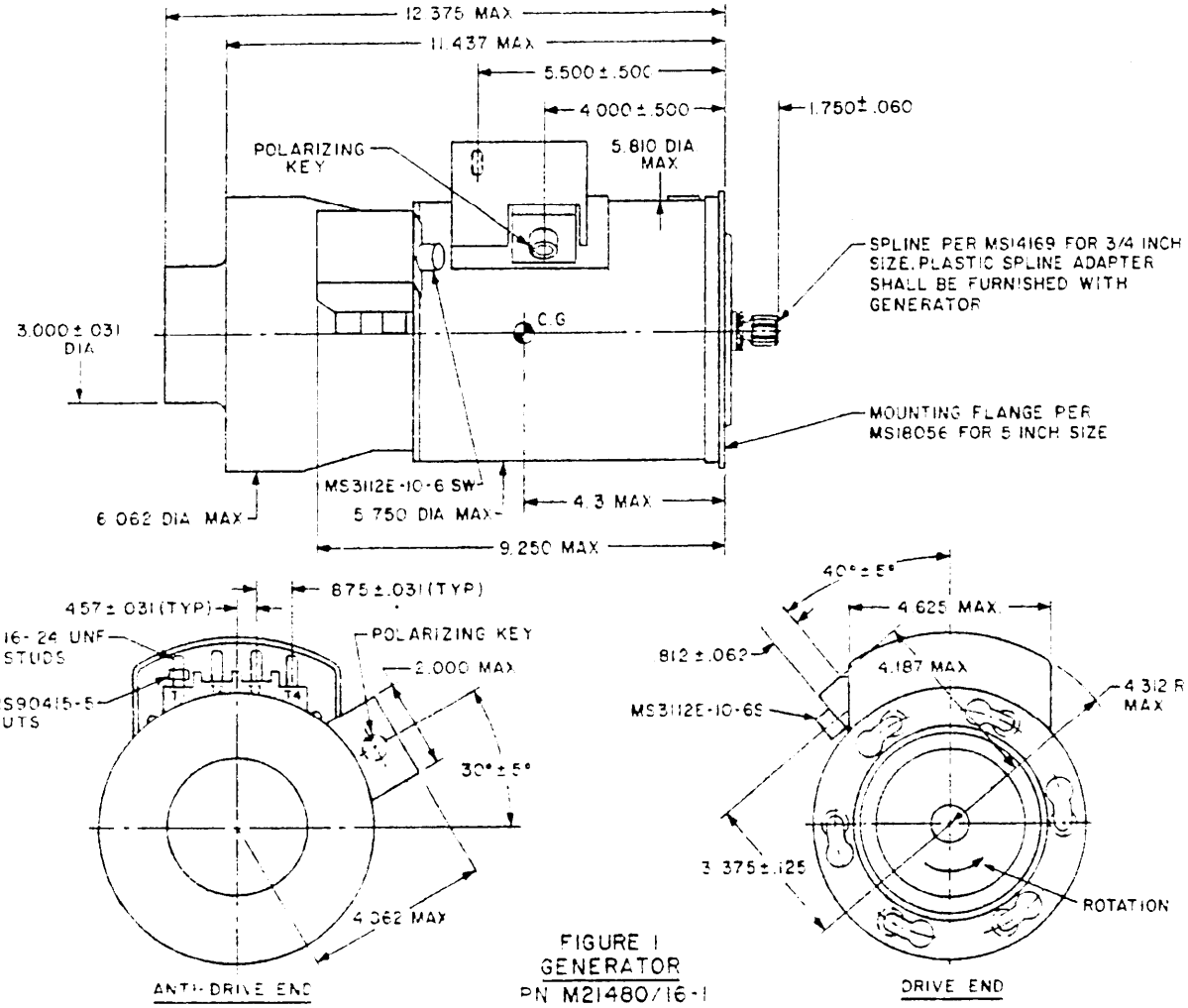


FIGURE 1
GENERATOR
PN M21480/16-1

GENERATOR DATA	
RATED VOLTAGE	120/208 VOLT
RATED OUTPUT	20 KVA
PHASE	3
FREQUENCY	380 TO 420 HZ
SPEED RANGE	11,400 TO 12,600 RPM
MAX SPEED FOR REGULATION	13,000 RPM
OVERSPEED	15,000 RPM
RATED POWER FACTOR	.75 LAG TO 0
EFFICIENCY MIN AT RATED LOAD	85%
MAXIMUM WEIGHT	23.5 POUNDS
OVERHUNG MOMENT	125 IN LBS (MAX)
SHAFT SHEAR SECTION	830-940 IN LBS
SELF-COOLING	YES
BEARING GREASE	MIL-G-6.322

FSC 6115

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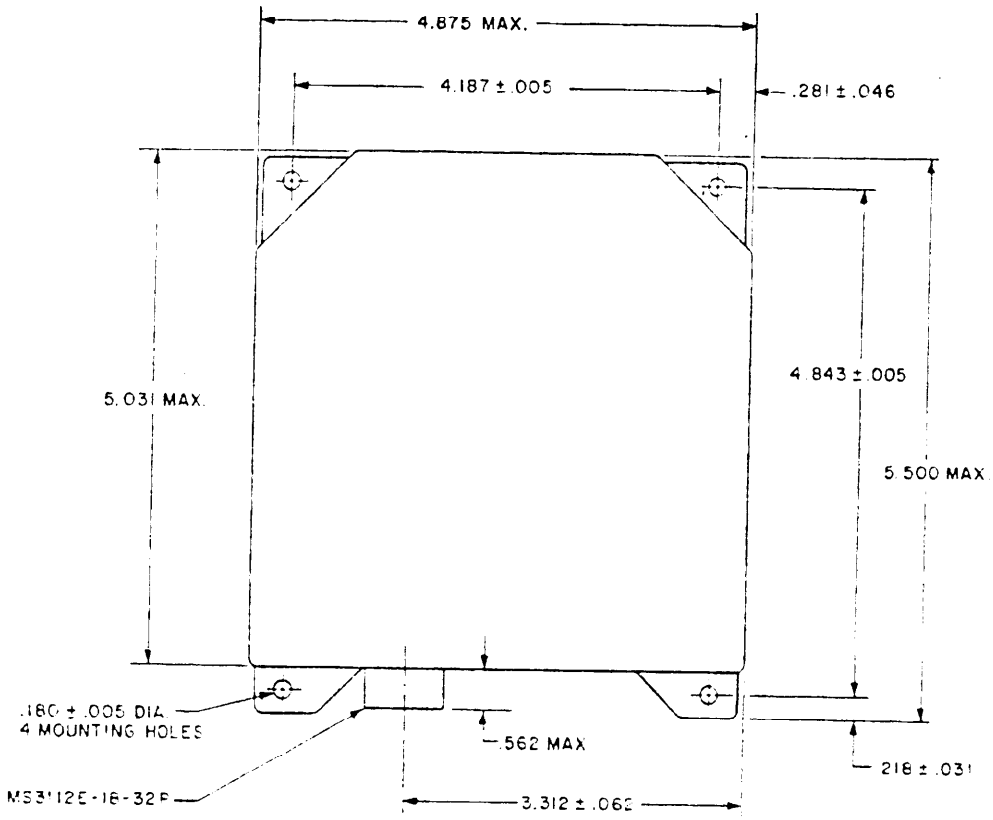


FIGURE 2
GENERATOR CONTROL UNIT, WT 3 LBS MAX.
PN M21480/16-4

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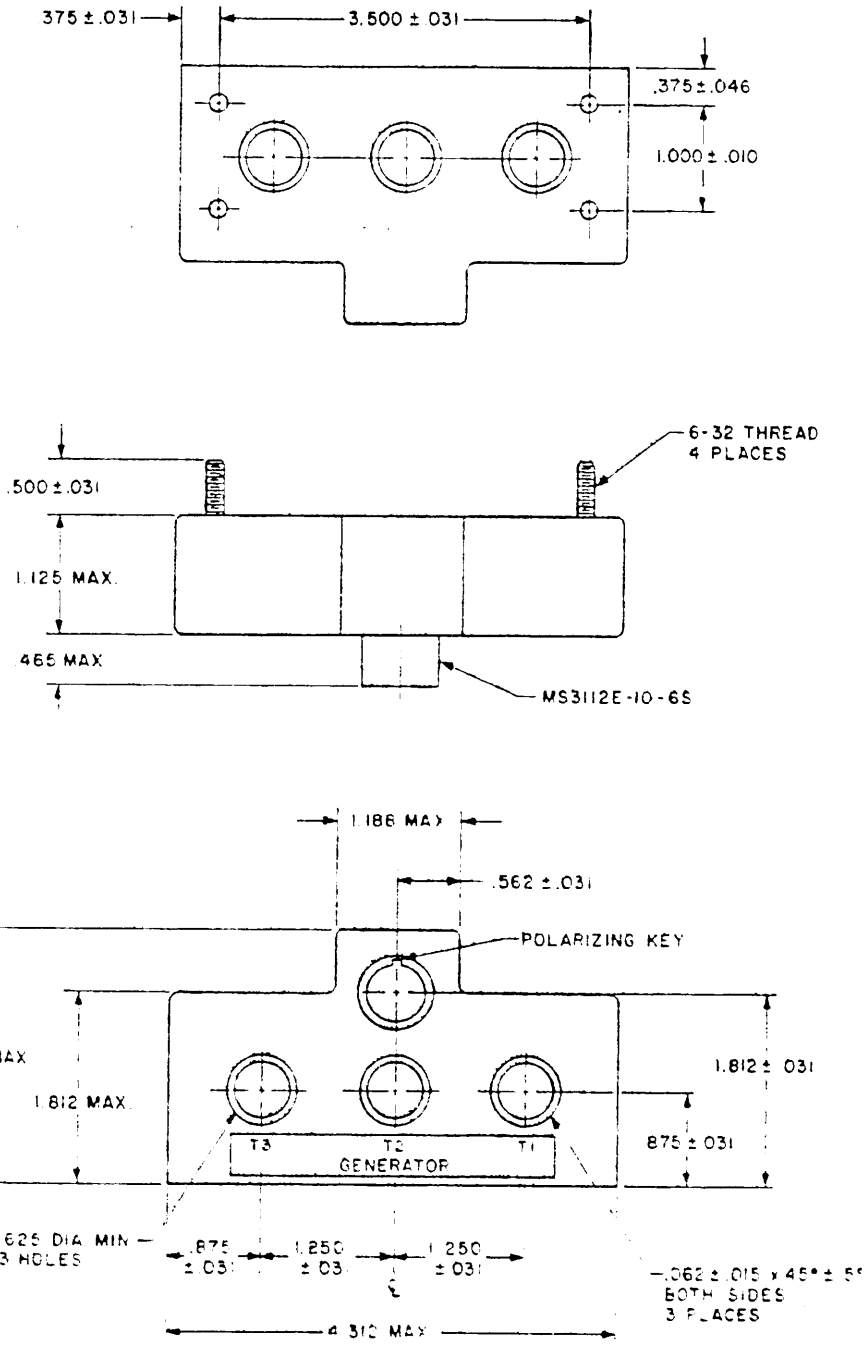


FIGURE 3
CURRENT TRANSFORMER, WT 0.75 LBS. MAX
PN M21480/16-5

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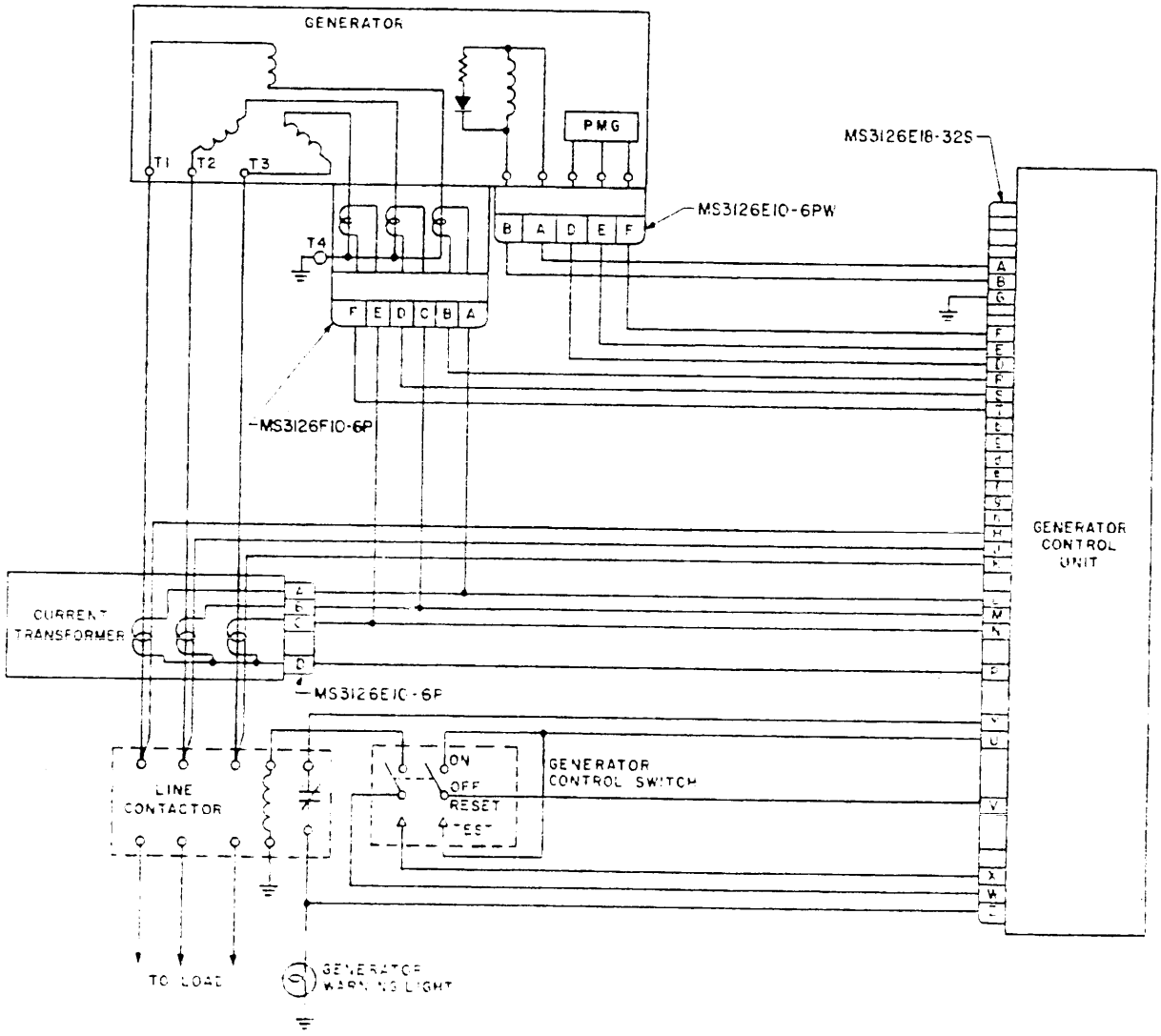


FIGURE 4
TEST CIRCUIT

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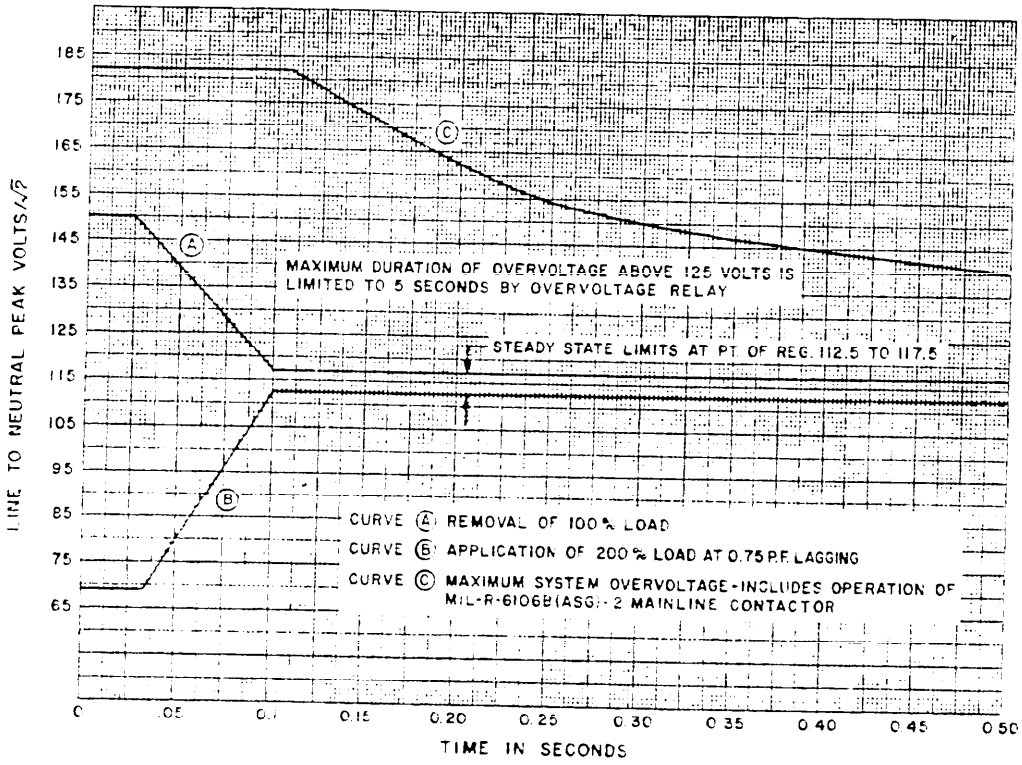


FIGURE 5
STEADY STATE, TRANSIENT AND OVERVOLTAGE LIMITS

REQUIREMENTS

THE REQUIREMENTS OF MIL-G-21480 SHALL APPLY WITH THE FOLLOWING EXCEPTIONS TO EXISTING PARAGRAPHS:

DELETE THE FOLLOWING PARAGRAPHS:

3.4.2.2 THROUGH 3.4.2.6, 3.4.10, 3.4.10.5, 3.4.10.6, 3.4.13, 3.4.13.1, 3.4.15, 3.5.7.5, 4.5.20, 4.5.22.1 THROUGH 4.5.22.3, 4.5.22.3.1 THROUGH 4.5.22.3.4, 4.5.24.1 THROUGH 4.5.24.3, 4.5.24.3.1 THROUGH 4.5.24.3.4, 4.5.25.1, 4.5.25.2, 4.5.25.2.1, 4.5.32, 4.5.33.1, AND 4.5.33.2.

3.4.6 ADD: THE FEEDERS FROM THE GENERATOR OUTPUT TERMINALS TO THE LINE CONTACTOR SHALL BE 6 AWG, 6.5 ± 0.5 FEET IN LENGTH. CHANGE "FIGURE 3" TO "FIGURE 5" OF THIS SPECIFICATION.

3.4.6.2 CHANGE "5 MINUTES" TO "2 MINUTES".

3.4.9.3 DELETE AND ADD: GENERATOR SHALL BE SUPPLIED WITH TERMINALS AS SHOWN IN FIGURE 1 OF THE DETAILED SPECIFICATION.

3.4.9.6.1.1 CHANGE "10 000 HOUR LIFE" TO "1000 HOUR LIFE".

3.4.11 CHANGE "ALL SYSTEM COMPONENTS" TO "GENERATOR AND GENERATOR CONTROL UNIT".

3.5.1 CHANGE "FIGURE 3" TO "FIGURE 5" OF THIS SPECIFICATION.

3.5.4 DELETE AND ADD: THE CREST FACTOR FOR EACH PHASE VOLTAGE WAVEFORM SHALL BE 1.41 ± 10%. NO SINGLE HARMONIC SHALL EXCEED 3% OF THE FUNDAMENTAL.

3.5.5 CHANGE "5 AMPS MINIMUM CONTINUOUS CAPACITY" TO "2 AMPS CONTINUOUS CAPACITY".

3.5.7.2 CHANGE "FIGURE 6" TO "FIGURE 5" OF THIS SPECIFICATION.

3.5.7.3 DELETE AND ADD: THE UNDERVOLTAGE PROTECTION DEVICE SHALL FUNCTION TO DISCONNECT THE SYSTEM FROM THE LOAD BUS WHEN ANY ONE PHASE VOLTAGE DROPS TO A VALUE OF 90 - 105 VOLTS FOR A TIME DELAY OF 4 ± 1 SECONDS. THE SYSTEM SHALL BE CAPABLE OF BEING RESET WHEN ALL THREE PHASE VOLTAGES ARE ABOVE DROP-OUT LEVEL AND BEFORE ANY PHASE EXCEEDS 105 VOLTS.

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3.5.7.4 DELETE AND ADD: THE UNDERFREQUENCY PROTECTION DEVICE SHALL FUNCTION TO DISCONNECT THE SYSTEM FROM THE LOAD BUS WHEN THE FREQUENCY DROPS TO A VALUE OF 370-380 HZ FOR A TIME DELAY OF 1 ± 0.5 SECOND. PICKUP VALUE SHALL BE GREATER THAN DROP-OUT LEVEL AND BEFORE THE FREQUENCY EXCEEDS 380 HZ.

3.5.7.6 CHANGE "25 AMPERES" TO "21 AMPERES".

3.5.8 DELETE AND ADD: THE SYSTEM SHALL MEET THE REQUIREMENTS OF MIL-E-81910, CLASS NO. III B FOR THE GENERATORS AND CONTROL EQUIPMENT, EXCEPT THE TEN MICROFARAD FEED-THROUGH CAPACITOR SHALL BE REMOVED.

4.2.1.1 ADD: QUALIFICATION TESTS SHALL BE SUPPLEMENTED WITH A MINIMUM 100 HOUR FAILURE FREE FLIGHT TEST BEFORE CONSIDERATION IS GIVEN TO INCORPORATING THE ITEM ON THE QUALIFIED PRODUCTS LIST.

4.5.3.1 CHANGE "FIGURE 3" TO "FIGURE 5" OF THIS SPECIFICATION.

4.5.3.2 DELETE AND ADD: EACH GENERATOR CONTROL UNIT SUBMITTED FOR ACCEPTANCE SHALL BE TESTED FOR VOLTAGE REGULATION AT 400 HZ, NO LOAD, FULL, 150%, AND 200% RATED LOADS. REPEAT AT 380 HZ AND 420 HZ AT NO LOAD AND FULL LOAD. VOLTAGES SHALL CONFORM TO THE LIMITS OF 115 ± 2.5 VOLTS AND VOLTAGE MODULATION FROM NO LOAD TO FULL LOAD SHALL NOT EXCEED 0.75%.

4.5.6 CHANGE "AN ACCEPTANCE OR QUALIFICATION TEST" TO "A QUALIFICATION TEST. THIS TEST IS NOT REQUIRED FOR ACCEPTANCE TESTING".

4.5.12 DELETE AND ADD: ELECTROMAGNETIC INTERFERENCE SHALL BE CONDUCTED AS SPECIFIED IN 3.5.8.

4.5.13 DELETE LAST THREE SENTENCES AND ADD: THE WAVEFORM SHALL BE WITHIN THE LIMITS SPECIFIED IN 3.5.4.

4.5.20.1 DELETE AND ADD: THE GENERATOR CONTROL UNIT AND CURRENT TRANSFORMER SHALL BE SUBJECTED TO A SALT FOG TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILD UP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF FULL AND 200% RATED LOADS, AND PROTECTIVE FUNCTIONS.

4.5.20.2 DELETE AND ADD: THE GENERATOR SHALL BE SUBJECTED TO AN OIL-SALT WATER INGESTION TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILD UP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF FULL AND 200% RATED LOADS, AND PROTECTIVE FUNCTIONS.

4.5.21 DELETE AND ADD: THE GENERATOR, GENERATOR CONTROL UNIT AND CURRENT TRANSFORMER SHALL BE SUBJECTED TO A FUNGUS TEST IN ACCORDANCE WITH MIL-E-81910. WITH THE INSTALLATION INSTRUCTIONS AND MOUNTING NUTS (AS REQUIRED BY PARAGRAPH 4.3.1) ATTACHED. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILD UP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF FULL AND 200% RATED LOADS, AND PROTECTIVE FUNCTIONS. MOUNTING NUTS ARE TO BE UNAFFECTED AND THE PACKAGE HOLDING THE NUTS SUITABLE FOR SHIPMENT. INSTRUCTIONS ARE TO BE LEGIBLE.

4.5.22 DELETE AND ADD: THE GENERATOR SHALL BE SUBJECTED TO A SAND AND DUST TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILD UP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF FULL AND 200% RATED LOADS, AND PROTECTIVE FUNCTIONS.

4.5.23.1.1 DELETE AND ADD: THE GENERATOR SHALL BE SUBJECTED TO A VIBRATION TEST IN ACCORDANCE WITH MIL-E-81910, EXCEPT THAT THE VIBRATION RANGE SHALL BE FROM 5 TO 500 HZ AND THE ANTI-DRIVE END OF THE GENERATOR SHALL BE LIMITED TO 25G'S. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILD UP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF FULL AND 200% RATED LOADS, AND PROTECTIVE FUNCTIONS.

4.5.23.1.2 DELETE AND ADD: THE GENERATOR CONTROL UNIT SHALL BE SUBJECTED TO A VIBRATION TEST IN ACCORDANCE WITH MIL-E-81910, EXCEPT THAT THE VIBRATION RANGE SHALL BE FROM 5 TO 500 HZ. AT THE COMPLETION OF THIS TEST, THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILD UP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF FULL AND 200% RATED LOADS, AND PROTECTIVE FUNCTIONS.

4.5.23.2 DELETE AND ADD: NO VIBRATION TEST IS REQUIRED FOR GENERATOR OR GENERATOR CONTROL UNIT AT TIME OF ACCEPTANCE.

4.5.24 DELETE AND ADD: THE GENERATOR, GENERATOR CONTROL UNIT AND CURRENT TRANSFORMER SHALL BE SUBJECTED TO A SHOCK TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF PROCEDURE I, THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILD UP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF FULL AND 200% RATED LOADS, AND PROTECTIVE FUNCTIONS. THERE SHALL BE NO FAILURE OF THE MOUNTING ATTACHMENTS AND THE TEST ITEM SHALL REMAIN IN PLACE AND NOT CREATE A HAZARD DURING PROCEDURE III.

4.5.25 DELETE AND ADD: THE GENERATOR, GENERATOR CONTROL UNIT, AND CURRENT TRANSFORMER, EXCEPT THOSE TESTED UNDER PARAGRAPH 4.5.20.2 SHALL BE SUBJECTED TO A HUMIDITY TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILD UP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF FULL AND 200% RATED LOADS, AND PROTECTIVE FUNCTIONS.

4.5.27 DELETE FIRST SENTENCE AND ADD: ALL SYSTEM COMPONENTS EXCEPT THE GENERATOR SHALL BE SUBJECTED TO AN ACCELERATION TEST IN ACCORDANCE WITH MIL-E-81910. AT THE COMPLETION OF THIS TEST THE UNIT SHALL SUCCESSFULLY DEMONSTRATE ITS BUILD UP CHARACTERISTICS, THE EFFECT OF APPLICATION AND REMOVAL OF FULL AND 200% RATED LOADS, AND PROTECTIVE FUNCTIONS.

NOTES:

1. DIMENSIONS IN INCHES.

ITEM	PART NUMBER
COMPLETE SYSTEM (CONSISTS OF ONE GENERATOR, ONE GENERATOR CONTROL UNIT, ONE CURRENT TRANSFORMER)	M21480/16
GENERATOR	M21480/16-1
GENERATOR CONTROL UNIT	M21480/16-4
CURRENT TRANSFORMER	M21480/16-5

FOR DESIGN FEATURE PURPOSES, THIS SPECIFICATION TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCE DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS.

PREPARING ACTIVITY
NAVY - AS
PROJECT NO. 6115-N379

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